

MODULE SPECIFICATION

Module Title (English)	Machine Learning for Science
Module Title (Chinese)	科学中的机器学习
Module Code	SCI402
Originating Department/School	School of Science
Module Level¹	Level 4 (FHEQ level 7)
Module Credits	5
Pre-requisites	N/A
Shared Programme(s)	MRes Advanced Chemical Sciences MRes Bioinformatics MRes Environmental Sciences MRes Molecular Bioscience MRes Materials Science and Engineering

Mode of Delivery and Hours²

	Lectures	Seminars	Tutorials	Lab / Practicals	Fieldwork / Placement	Other (Private study)	Total
Hours / Semester or Year³	20		6	30		94	150
Delivery Pattern	10*1*2		3*1*2	6*1*5			

Description

Aims and Fit of Module

In the science field, machine learning offers a powerful set of tools for data analysis, pattern recognition, and predictive modelling. It allows scientists to handle large and complex datasets, extract meaningful insights, and make data-driven decisions. This module aims to provide scientists and researchers with a comprehensive understanding of machine learning techniques and their applications in scientific research.

By integrating machine learning into science programs, students gain the ability to transform their raw data into actionable knowledge, which can lead to new discoveries, more efficient processes, and better-informed policies. The interdisciplinary nature of machine learning ensures that it is a valuable addition to any science curriculum, preparing students for the data-rich future of scientific research and application.

Learning Outcomes

Students completing the module successfully should be able to:

A. Pre-process data by handling missing values, scaling, normalization, and feature selection and engineering, which is essential for feeding data into machine learning models.

B. Define machine learning, differentiate between supervised and unsupervised learning, and explain the concept of overfitting and underfitting.

C. Apply and understand the strengths and weaknesses of machine learning algorithms such as linear regression, decision trees, random forests, support vector machines, k-nearest neighbours, and neural networks.

D. Evaluate the performance of machine learning models using various metrics and how to select the appropriate model based on the data and the scientific problem at hand.

E. Apply machine learning techniques to real-world scientific problems, understand the limitations and challenges of machine learning in scientific research, and be equipped with the skills to continue learning and adapting to new developments in the field.

Method of Teaching and Learning

Each lecture in the module is complemented by hands-on Python demonstrations, bridging theory and practice. Python's extensive libraries make it ideal for machine learning, enabling students to grasp complex concepts through coding exercises. Students will work with high-quality scientific datasets to apply AI to real-world problems, gaining practical experience in solving real-life challenges.

The module includes tutorial sessions to reinforce lecture content. These interactive sessions provide additional explanations, answer questions, and guide problem-solving exercises, helping students deepen their understanding of complex topics.

Multiple lab sessions throughout the module expose students to diverse case studies and examples, focusing on applying AI and machine learning techniques to scientific data analysis. These sessions are crucial for developing practical skills in data preprocessing, model building, evaluation, and interpretation. Students will experiment with different techniques and tools, fostering creativity and problem-solving abilities.

Overall, this approach ensures a comprehensive understanding of machine learning for science. Combining lectures, demonstrations, tutorials, and lab sessions, students will develop both theoretical knowledge and practical skills, preparing them to tackle complex scientific problems using machine learning.

Syllabus

Topic 1: Introduction to Machine Learning
 Topic 2: Data Pre-processing
 Topic 3: Supervised Learning
 Topic 4: Unsupervised Learning
 Topic 5: Weakly supervised learning
 Topic 6: Reinforcement learning

Assessment

Initial Assessment

Sequence	Assessment Type(EXAM or CW)	Method	Learning outcomes assessed(<i>use codes under Learning Outcomes</i>)	Duration ⁴	% of Final Mark	Resit(Y/N/S) ⁵
001	CW	Essay	ABC		15	S
002	CW	Project (Proposal)	ABCD		25	S

		Development)				
003	CW	Project	ABCDE		60	S

Resit Assessment

Assessment Type (EXAM or CW)⁶	Learning outcomes assessed (use codes under Learning Outcomes)	Duration	% of Final Mark
CW	ABCDE		100

The resit assessment will assess all of the learning outcomes of the module, and will be weighted as 100% of the final module mark. Other components of the assessment, regardless of whether or not the student passed or failed, will not be included in the calculation of the final module mark, following resit assessment.

Textbooks

Mandatory Textbooks

Title	Author	ISBN/Publisher
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Optional Textbooks

Title	Author	ISBN/Publisher
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Reference Textbooks

Title	Author	ISBN/Publisher
MACHINE LEARNING: A PROBABILISTIC PERSPECTIVE	KEVIN P. MURPHY	9780262044660 /THE MIT PRESS

¹This indicates both XJTLU module level and FHEQ level (The Frameworks for HE Qualifications of UK Degree-Awarding Bodies: <https://www.qaa.ac.uk/quality-code/qualifications-and-credit-frameworks>).

²Total hours equal credits×30.

³For year-long modules, this refers to the total hours for the year.

⁴This column specifies the duration of the EXAM.

⁵Resit Codes:

Y: Retake the same assessment type with identical weighting

S: Single resit covering all learning outcomes, worth 100% of the final mark

N: No resit opportunity

⁶Either "exam" or "coursework" is selected here to indicate the type of the resit assessment.